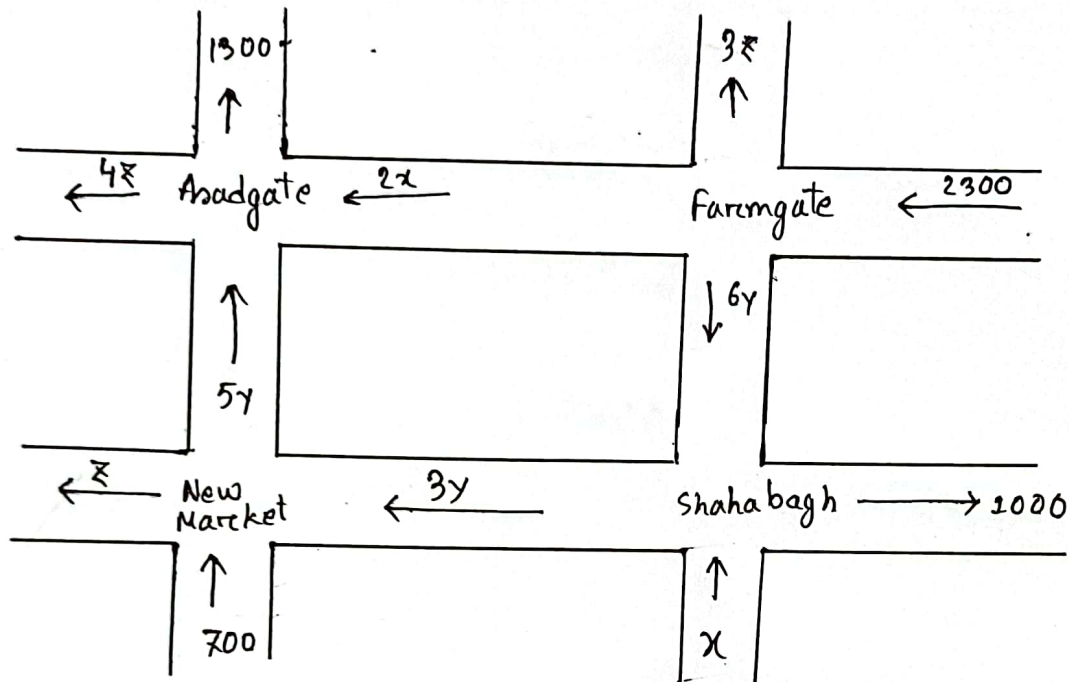


The following diagram describes the traffic flow of a part of congested part of Dhaka on some day.



(a) Form the system of linear equations.

(b) Solve the system.

Solution: Summarizing the information, we get

<u>Intersection</u>	<u>Incoming traffic</u>	<u>Outgoing traffic</u>
Asadgate	$2x + 5y$	$4z + 1300$
New market	$3y + 700$	$5y + z$
Farm gate	2300	$2x + 6y + 3z$
Shahbagh	$x + 6y$	$3y + 1000$

At each intersection, the incoming traffic must match the outgoing traffic so that there will be no congestion.

Therefore we have,

$$2x + 5y = 4z + 1300$$

$$3y + 700 = 5y + z$$

$$2300 = 2x + 6y + 3z$$

$$x + 6y = 3y + 1000$$

So, the system of linear equation is-

$$2x + 5y - 4z = 1300$$

$$2y + z = 700$$

$$2x + 6y + 3z = 2300$$

$$x + 3y = 1000$$

-①

The system can be written as,

$$AX=B$$

where,

$$A = \begin{pmatrix} 2 & 5 & -4 \\ 0 & 2 & 1 \\ 2 & 6 & 3 \\ 1 & 3 & 0 \end{pmatrix}, X = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \text{ and } B = \begin{pmatrix} 1300 \\ 700 \\ 2300 \\ 1000 \end{pmatrix}$$

The augmented matrix of the system is

$$[A|B] = \left(\begin{array}{ccc|c} 2 & 5 & -4 & 1300 \\ 0 & 2 & 1 & 700 \\ 2 & 6 & 3 & 2300 \\ 1 & 3 & 0 & 1000 \end{array} \right)$$

$$\sim \left(\begin{array}{ccc|c} 2 & 5 & -4 & 1300 \\ 0 & 2 & 1 & 700 \\ 0 & 1 & 7 & 1000 \\ 1 & 3 & 0 & 1000 \end{array} \right) \quad R_3' = R_3 - R_1$$

$$\sim \left(\begin{array}{ccc|c} 2 & 5 & -4 & 1300 \\ 0 & 2 & 1 & 700 \\ 0 & 1 & 7 & 1000 \\ 0 & 1 & 4 & 700 \end{array} \right) \quad R_4' = 2R_4 - R_1$$

$$\sim \left(\begin{array}{ccc|c} 2 & 5 & -4 & 1300 \\ 0 & 2 & 1 & 700 \\ 0 & 0 & 3 & 300 \\ 0 & 1 & 4 & 700 \end{array} \right) \quad R_3 = R_3 - R_4$$

$$\sim \left(\begin{array}{ccc|c} 2 & 5 & -4 & 1300 \\ 0 & 2 & 1 & 700 \\ 0 & 0 & 3 & 300 \\ 0 & 0 & 7 & 700 \end{array} \right) \quad R_4' = 2R_4 - R_2$$

$$\sim \left(\begin{array}{ccc|c} 2 & 5 & -4 & 1300 \\ 0 & 2 & 1 & 700 \\ 0 & 0 & 3 & 300 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad R_4' = 3R_4 - 7R_3$$

which is the echelon form of the augmented matrix.

Now, $P(A|B) = P(A)$

So, the solution of the system exists.

And No. of variable = Rank

$$\Rightarrow 3 = 3$$

So, the system is consistent and has unique solution.

the reduced system is,

$$2x + 5y - 4z = 1300$$

$$2y + z = 700$$

$$3z = 300$$

By back substitution we get,

$$3z = 300 \Rightarrow z = 100$$

$$2y + z = 700 \Rightarrow 2y + 100 = 700 \Rightarrow y = 300$$

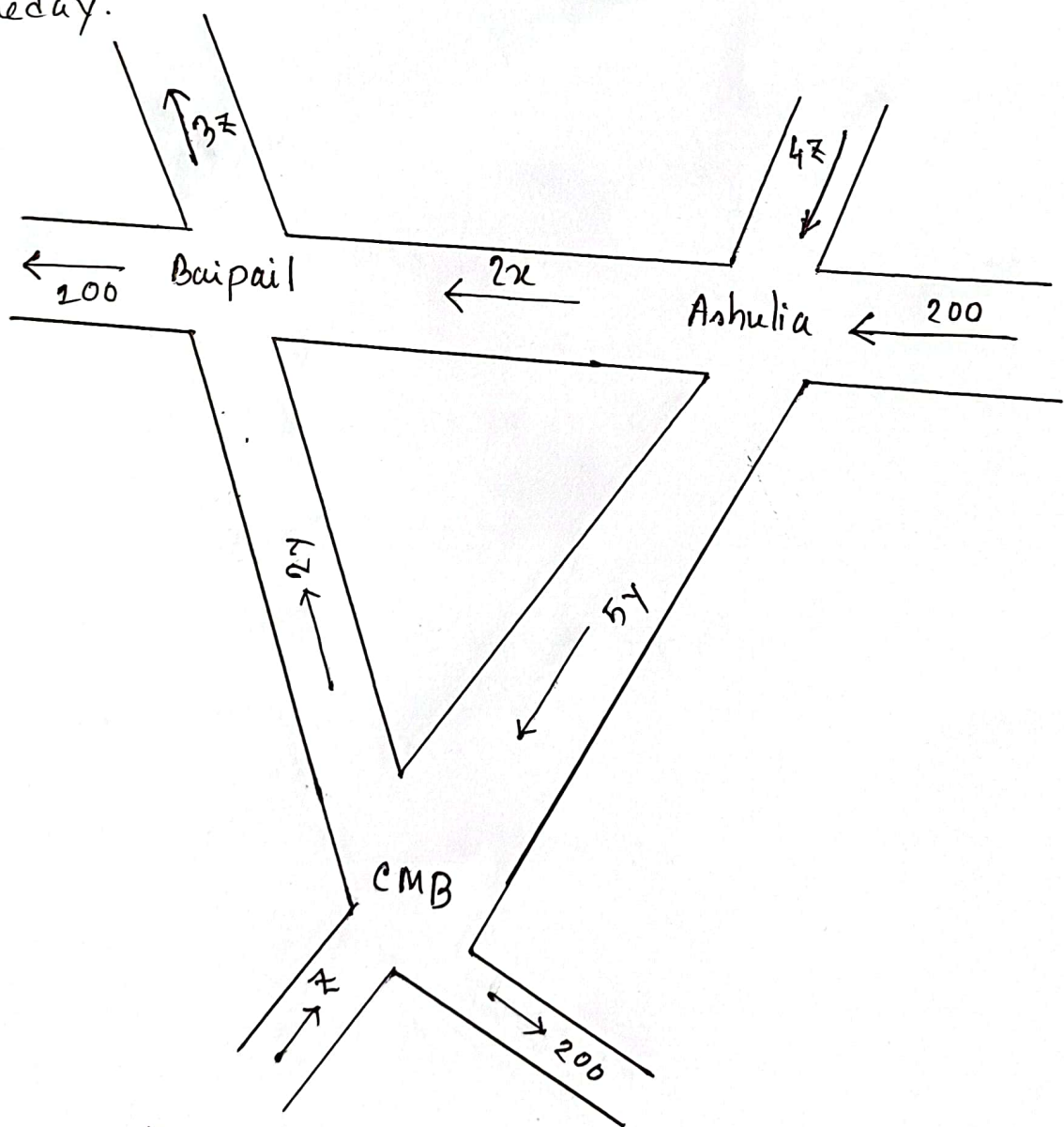
$$2x + 5y - 4z = 1300 \Rightarrow 2x + 5 \cdot 300 - 4 \cdot 100 = 1300 \Rightarrow x = 100$$

Hence the required result is,

$$(x, y, z) = (100, 300, 100)$$

$\therefore$ solution set, $= \{100, 300, 100\}$ Ans

The following diagram describes the traffic flow of a part of congested part of Savan Upazila on someday.



(a) Form the SLE

(b) Solve the system.

Solution: Summarizing the information, we get

<u>Intersection</u>	<u>Incoming traffic</u>	<u>Outgoing traffic</u>
Baipail	$2x + 2y$	$3z + 100$
Ashulia	$4z + 200$	$2x + 5y$
CMB	$5y + z$	$2y + 200$

At each intersection, the incoming traffic must match the outgoing traffic so that there'll be no congestion, therefore we have,

$$2x + 2y = 3z + 100$$

$$4z + 200 = 2x + 5y$$

$$5y + z = 2y + 200$$

So, the system of linear equation is,

$$\left. \begin{aligned} 2x + 2y - 3z &= 100 \\ 2x + 5y - 4z &= 200 \\ 3y + z &= 200 \end{aligned} \right\} \rightarrow \textcircled{1}$$

The system can be written as

$$AX = B$$

where, $A = \begin{pmatrix} 2 & 2 & -3 \\ 2 & 5 & -4 \\ 0 & 3 & 1 \end{pmatrix}$ $x = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ And $B = \begin{pmatrix} 100 \\ 200 \\ 200 \end{pmatrix}$

The Augmented matrix of the system is,

$$[A|B] = \left(\begin{array}{ccc|c} 2 & 2 & -3 & 100 \\ 2 & 5 & -4 & 200 \\ 0 & 3 & 1 & 200 \end{array} \right)$$

$$\sim \left(\begin{array}{ccc|c} 2 & 2 & -3 & 100 \\ 0 & 3 & -1 & 200 \\ 0 & 3 & 1 & 200 \end{array} \right) \quad R_2' = R_2 - R_1$$

$$\sim \left(\begin{array}{ccc|c} 2 & 2 & -3 & 100 \\ 0 & 3 & -1 & 100 \\ 0 & 0 & 2 & 100 \end{array} \right) \quad R_3' = R_3 - R_2$$

which is the echelon form of the augmented matrix.

$$\text{Now, } \rho(A|B) = \rho(A)$$

So, the solution of the linear equation exists.

And No. of variable = Rank

$$3 = 3$$

So, the system is consistent, has unique solution.

The reduced system is,

$$2z = 100 \Rightarrow z = 50$$

$$3y - z = 100 \Rightarrow 3y - 50 = 100 \Rightarrow y = 50$$

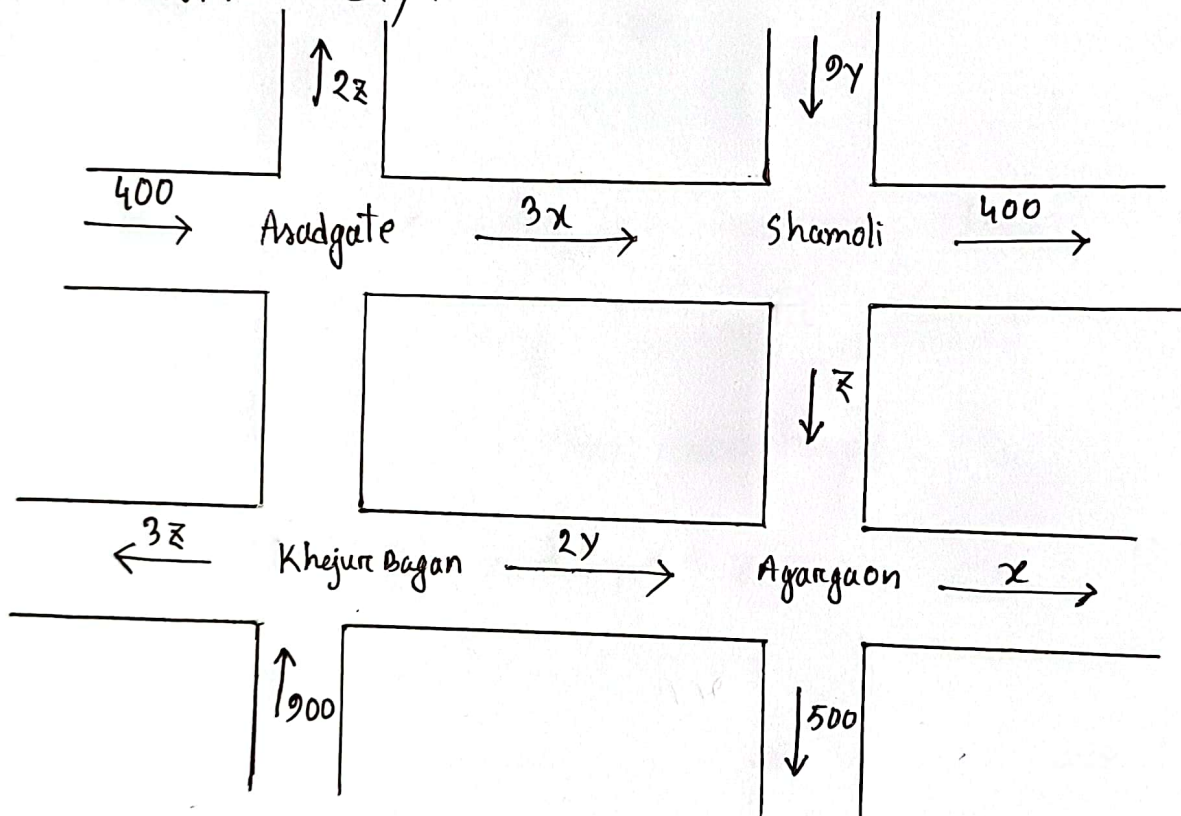
$$2x + 2y - 3z = 100 \Rightarrow 2x + 2 \cdot 50 - 3 \cdot 50 = 100 \Rightarrow x = 75$$

Hence, the required result is

$$(x, y, z) = (75, 50, 50)$$

$\therefore$ Solution set = $\{75, 50, 50\}$ Ans:

The following diagram describes the traffic flow of a part of congested part of Dhaka on some day.



(a) From the SLE

(b) Solve the system.

Solution: , Summarizing the information we get,

<u>Intersection</u>	<u>Incoming traffic</u>	<u>Outgoing traffic</u>
Asad gate	400	$3x + 3y + 2z$
Shamoli	$3x + 0y$	$z + 400$
Khejurbagan	$3y + 000$	$2y + 5z$
Agargaon	$2y + z$	$x + 500$

At each intersection, the incoming traffic must match the outgoing traffic so that there will be no congestion. Therefore we have,

$$400 = 3x + 3y + 2z$$

$$3x + 0y = 400 + z$$

$$2y + z = x + 500$$

So, the system of linear equation is,

$$3x + 3y + 2z = 400$$

$$3x + 0y - z = 400$$

$$-y + 5z = 000$$

$$-x + 2y + z = 500$$

} — ①

The system ① can be written as,

$$AX = B$$

where $A = \begin{pmatrix} 3 & 3 & 2 \\ 3 & 0 & -1 \\ 0 & -1 & 5 \\ -1 & 2 & 1 \end{pmatrix}$, $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ And $B = \begin{pmatrix} 400 \\ 400 \\ 900 \\ 500 \end{pmatrix}$

the augmented matrix of the system is,

$$\left(\begin{array}{ccc|c} 3 & 3 & 2 & 400 \\ 3 & 0 & -1 & 400 \\ 0 & -1 & 5 & 900 \\ -1 & 2 & 1 & 500 \end{array} \right)$$

$$\sim \left(\begin{array}{ccc|c} 3 & 3 & 2 & 400 \\ 0 & 6 & -3 & 0 \\ 0 & -1 & 5 & 900 \\ -1 & 2 & 1 & 500 \end{array} \right) \quad R_2' = R_2 - R_1$$

$$\sim \left(\begin{array}{ccc|c} 3 & 3 & 2 & 400 \\ 0 & 6 & -3 & 0 \\ 0 & -1 & 5 & 900 \\ 0 & 0 & 5 & 1900 \end{array} \right) \quad R_4' = 3R_4 + R_1$$

$$\sim \left(\begin{array}{ccc|c} 3 & 3 & 2 & 400 \\ 0 & 6 & -3 & 0 \\ 0 & 0 & 27 & 5400 \\ 0 & 0 & 5 & 1900 \end{array} \right) \quad R_3' = R_2 + 6R_3$$

$$\sim \left(\begin{array}{ccc|c} 3 & 3 & 2 & 400 \\ 0 & 6 & -3 & 0 \\ 0 & 0 & 27 & 5400 \\ 0 & 0 & 54 & 11400 \end{array} \right) \quad R_4' = 6R_4 - 9R_2$$

$$\sim \left(\begin{array}{ccc|c} 3 & 3 & 2 & 400 \\ 0 & 6 & -3 & 0 \\ 0 & 0 & 27 & 5400 \\ 0 & 0 & 0 & 0 \end{array} \right) \quad R_4' = 2R_4 - 57R_2$$

which is the echelon form of augmented matrix.

$$\text{Now, } \rho(A|B) = \rho(A)$$

And, No. of variable = Rank

$$3 = 3$$

So, the solution of the linear equation exists and the system is consistent and has unique solution.

The reduced system is,

$$3x + 3y + 2z = 400$$

$$6y - 3z = 0$$

$$27z = 5400$$

By back substitution we get,

$$27z = 5400$$

$$\therefore z = 200$$

$$6y - 3z = 0 \Rightarrow 6y - 3 \cdot 200 = 0 \Rightarrow y = 100$$

$$3x + 3y + 2z = 400 \Rightarrow 3x + 3 \cdot 100 + 2 \cdot 200 = 400 \Rightarrow x = -100$$

Hence the required result is

$$(x, y, z) = (-100, 100, 200)$$

$$\text{Solution Set} = \{-100, 100, 200\} \quad \text{Ans:}$$